Computers and Education: Artificial Intelligence 7 (2024) 100284

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Evaluating the psychometric properties of ChatGPT-generated questions

Shreya Bhandari [a], Yunting Liu [b], Yerin Kwak [b], Zachary A. Pardos [b,*]

[a] University of California, Berkeley, EECS, Berkeley, CA, USA
[b] University of California, Berkeley, School of Education, Berkeley, CA, USA

A R T I C L E I N F O

Keywords:
Formative assessment
Generative AI
Item response theory
Psychometric measurement
Large language models

A B S T R A C T

Not much is known about how LLM-generated questions compare to gold-standard, traditional formative assessments concerning their difficulty and discrimination parameters, which are valued properties in the psychometric measurement field. We follow a rigorous measurement methodology to compare a set of ChatGPT-generated questions, produced from one lesson summary in a textbook, to existing questions from a published Creative Commons textbook. To do this, we collected and analyzed responses from 207 test respondents who answered questions from both item pools and used a linking methodology to compare IRT properties between the two pools. We find that neither the difficulty nor discrimination parameters of the 15 items in each pool differ statistically significantly, with some evidence that the ChatGPT items were marginally better at differentiating different respondent abilities. The response time also does not differ significantly between the two sources of items. The ChatGPT-generated items showed evidence of unidimensionality and did not affect the unidimensionality of the original set of items when tested together. Finally, through a fine-grained learning objective labeling analysis, we found greater similarity in the learning objective distribution of ChatGPT-generated items and the items from the target OpenStax lesson (0.9666) than between ChatGPT-generated items and adjacent OpenStax lessons (0.6859 for the previous lesson and 0.6153 for the subsequent lesson). These results corroborate our conclusion that generative AI can produce algebra items of similar quality to existing textbook questions that share the same construct or constructs as those questions.

1. Introduction

Much research has been conducted on different processes of question generation (Sayin & Gierl, 2024), question difficulty control (Lu & Wang, 2024), and psychometric validation (Colvin et al., 2014; Liu et al., 2024). However, the current body of literature lacks the use of psychometric methods to evaluate the properties of AI-generated items (Tan et al., 2024). If high-quality question generation is possible with AI, we will be able to develop intelligent tutoring systems capable of both unlimited question generation and effective control of question difficulty. The potential problems this system would solve are multifaceted. Firstly, the lack of control over question difficulty poses issues such as generating questions of inappropriate difficulty that hinder student learning and requiring tedious manual question searches from exam designers. With an intelligent tutoring system that can efficiently control difficulty, these issues would be mitigated. Secondly, a system capable of unlimited question generation reduces the learning barrier caused by lack of practice and optimizes the labor-intensive process of generating exam questions. This methodology may also be applicable to designing summative tests; however, in this work, we only validate the generation against existing formative assessments used in practice.

Item development is a critical component of contemporary assessment systems, playing a pivotal role in evaluating knowledge and learning objectives (Irvine & Kyllonen, 2013). With the help of high-quality items, we are able to draw inferences about students' current knowledge states through different models (Wilson, 2013), thus facilitating better instruction. In a formative testing environment, effective assessment can afford practitioners the opportunity to diagnose subjects in a way that leads to tailored remediation. However, this process is associated with substantial costs in terms of both finances and time, often demanding extensive efforts from professionals within the relevant testing domain. In particular, traditional measurement practice requires each item to be written, edited, reviewed, and revised by subject matter experts (SMEs) until they meet the appropriate standard of quality (Lane et al., 2015). Therefore, if a large number of items is needed for the ITS system, a large number of SMEs must be involved, requiring significant effort (Sayin &

* Corresponding author.
E-mail address: pardos@berkeley.edu (Z.A. Pardos).

https://doi.org/10.1016/j.caeai.2024.100284
Received 31 December 2023; Received in revised form 5 August 2024; Accepted 17 August 2024
Available online 22 August 2024

Gierl, 2024). Consequently, transformative solutions are sought to cut down costs and enhance the production of effective assessments.

Large Language Models (LLMs), such as ChatGPT,[1] are bringing rise to speculation on whether they can competently produce math questions (Onal & Kulavuz-Onal, 2024; Kurdi et al., 2020). While some research has been done on other large language models, such as BART and T5, producing math questions (Nguyen et al., 2022; Shridhar et al., 2022; Zhou et al., 2023), rigorous IRT analyses have not been conducted, and little is currently known about the validity of ChatGPT-generated questions and how they compare to human-generated textbook questions. Therefore, in this work, we aim to compare the quality of ChatGPT and human-generated items to answer the following research questions:

- **RQ1:** What is the range of IRT item difficulties and discrimination parameters for ChatGPT-generated questions? How do they compare to human-authored questions?
- **RQ2:** What is the range of response times for ChatGPT-generated questions? How do they compare to human-authored questions?
- **RQ3:** Do ChatGPT-generated questions for each lesson conform to the dimensionality assumption based on item responses and NLP metrics?

We make all of the item content used in the study available under a creative commons license[2] and provide the source code and a standing demo of the system used to collect respondent data[3] allowing for full replication of our study environment.

2. Related work

Generative AI using ChatGPT is seeing nascent applications in education. It has been used to generate help messages and other types of feedback in Algebra (Pardos & Bhandari, 2023) and in a decimal learning game setting (Nguyen et al., 2023). ChatGPT has also been used for evaluative purposes, assessing the quality of crowd-sourced multiple-choice questions using a rubric included as part of the prompt (Moore et al., 2023). In these examples, the efficacy of the technology was promising but not without reported shortcomings. In the remainder of the related work section, we focus on work related to question generation, textbook evaluation, and LLMs in general.

2.1. LLMs and question generation

Large Language Models (LLMs) are being pre-trained and fine-tuned with ever-larger text corpora, giving rise to highly advanced and domain-general models. These models typically leverage foundation model architectures (Bommasani et al., 2021) associated with the Transformer (Vaswani et al., 2017) and its novel attention mechanism. GPT (Radford et al., 2018) is a notable example that uses the Transformer as its base architecture, specifically utilizing decoding components (i.e., generating-oriented). ChatGPT is an interface to a machine learning model based on the Generative Pre-trained Transformer (GPT) architecture and fine-tuned using Reinforcement Learning with Human Feedback (Griffith et al., 2013). The GPT model has undergone years of development through autoregressive language modeling and has experienced several stages of evolution. Most recently, the model has utilized human raters to fine-tune it to better respond to common human prompts and provide desirable responses (Radford et al., 2018, 2019; Brown et al., 2020; Long Ouyang et al., 2022). With ChatGPT being based on such an advanced model, curiosity has arisen regarding the extent of its capabilities.

The majority of past research in this field has focused on leveraging the capabilities of LLMs to create math questions based on specifying a template-based approach (Shridhar et al., 2022), open-ended generation (Socratic style math questions or math word problems) (Shridhar et al., 2022; Zhou et al., 2023; Keller, 2021; Onal & Kulavuz-Onal, 2024), or multiple-choice question generation (Dhawaleswar & Saha, 2023). The likely reason these approaches were taken, rather than simply having the LLMs emulate the creation of textbook questions, is that prior language model approaches to the prompt/response scenario treated the response as a text completion of the prompt. However, users tend to interact differently with these language models. They prefer to pose a question or give an instruction rather than provide a simple text completion. This misalignment motivated the creation of InstructGPT (or GPT-3.5) (Long Ouyang et al., 2022), the basis for ChatGPT. With the advent of ChatGPT providing this alignment, we aim to test the capabilities of ChatGPT for textbook question generation, given solely the textbook content. In our study, we utilize GPT-4, the March 14, 2023 version of the model, to produce questions.

2.2. Formative assessment evaluation

Formative assessment items differ from traditional summative assessments in that summative test results are not used for high-stakes purposes, such as school admissions criteria, but are instead used as part of instruction during the learning process. Some research argues that formative assessments should facilitate active learning and engagement as a vital factor (Overman et al., 2013), while others argue from a content perspective, stating that it is important for formative assessment items to encompass higher-level learning objectives or show multidimensionality (Damanik & Zainil, 2019; Mishra, 2015). The content perspective is also stressed in the formative assessment literature (Sireci & Zenisky, 2011), which states that an important feature of current formative assessments is better coverage of the construct being measured, termed 'Representativeness.' In summary, an optimal set of formative assessment items for a given unit of instruction (e.g., a textbook chapter) should have the following features: a) They should have good content coverage of the construct they are measuring, ideally consisting of some higher-level construct that may be a progression of the current one (Wilson, 2009). b) They should facilitate student thinking and subsequent studying, which means they should be interesting, probably innovative, and of moderate difficulty so as not to frighten students. Because the questions found in tutoring systems are also formative assessments, questions generated to satisfy the above criteria may be beneficial in both tutoring and textbook contexts.

2.3. Psychometric linking/equating

Assessment designers often have more items they would like to evaluate than are reasonable for respondents to answer on a single test. To split up the number of items on each test while retaining the ability to compare them to one another, measurement researchers utilize a technique called psychometric linking/equating to map different calibration results onto a common scale, thus ensuring parameters are comparable across multiple respondent test phases (van der Linden & Barrett, 2016; Wright & Stone, 1979). There are a number of ways linking can be accomplished, yet all methods involve an anchoring design and a linking transformation. Anchors can be either persons who answer items from both tests or items answered by different respondent groups (Vale, 1986). In a common item design, a subset of the items is included in both tests. The parameters of these common items are compared to determine the linear transformation necessary to put all items onto a common scale. This approach has been shown to be effective across the fields of Education (von Davier et al., 2019), Medical Health (McHorney, 2002), and Psychology (Joo et al., 2017). In 2013, Meyer and Zhu introduced linking methods for assessments on the online higher education platform MOOC (Meyer & Zhu, 2013).

[1] https://chat.openai.com/chat.
[2] https://github.com/CAHLR/OATutor-Question-Generation-Final.
[3] https://cahlr.github.io/OATutor-Question-Generation-Final.

2.2 Linear Equations in One Variable

- We can solve linear equations in one variable in the form $ax + b = 0$ using standard algebraic properties. See Example 1 and Example 2.
- A rational expression is a quotient of two polynomials. We use the LCD to clear the fractions from an equation. See Example 3 and Example 4.
- All solutions to a rational equation should be verified within the original equation to avoid an undefined term, or zero in the denominator. See Example 5 and Example 6 and Example 7.
- Given two points, we can find the slope of a line using the slope formula. See Example 8.
- We can identify the slope and y-intercept of an equation in slope-intercept form. See Example 9.
- We can find the equation of a line given the slope and a point. See Example 10.
- We can also find the equation of a line given two points. Find the slope and use the point-slope formula. See Example 11.
- The standard form of a line has no fractions. See Example 12.
- Horizontal lines have a slope of zero and are defined as $y = c$, where c is a constant.
- Vertical lines have an undefined slope (zero in the denominator), and are defined as $x = c$, where c is a constant. See Example 13.
- Parallel lines have the same slope and different y-intercepts. See Example 14 and Example 15.
- Perpendicular lines have slopes that are negative reciprocals of each other unless one is horizontal and the other is vertical. See Example 16.

Fig. 1. OpenStax Chapter 2.2 Key Concepts.

3. Methods

3.1. Subject selection and item generation

College Algebra was selected as the subject area because it is often an important bridging material at many colleges. It is also a subject for which pre-authored questions were available under a CC BY license from a notable open textbook publisher, OpenStax[4] (Abramson, 2021). To decide which lesson would be utilized in the study, we looked for the first lesson in which none of the problems depended on any images or figures, since a limitation of ChatGPT and most other LLMs, as of this writing, is that they are not yet multi-modal in their inputs and outputs. The first lesson that satisfied this criterion was *Lesson 2.2: Linear Equations in One Variable*. We selected 15 OpenStax questions from that lesson to utilize in our study. This number was chosen to “right-size” the test given to respondents. We used a random number generator to randomly select 15 questions from the pool of available items in that lesson. At the end of each chapter, there was a “Chapter Review” with “Key Concepts,” summarizing each lesson with bullet points (Fig. 1). We leveraged this when prompting GPT-4. To prompt ChatGPT to generate questions, the following prompt was given to ChatGPT:

< Period delimited list from Key Concepts >

Please generate 20 exercise questions based on this textbook chapter.

Each question was quality checked against the following 3-point criteria:

- Use of inappropriate language
- The question is solvable
- The question leads to a single solution (i.e. does not have multiple answers)

Each question generated by ChatGPT was manually attempted to be solved by the first author of the paper to ensure that the question was solvable. If a question failed one or more of the quality checks, it was eliminated from our question pool. Among the questions that passed all the checks, we used a random number generator to choose 15 questions to be included in our study. We did not apply a similar quality check to the OpenStax questions, as it is assumed that a separate quality control process was used internally by the publisher before the items were made public in their textbook.

As an example, the first question generated by ChatGPT was “Solve for x: $5x + 10 = 0$,” which corresponds to the first bullet point in the OpenStax Chapter 2.2 Key Concepts (Fig. 1).

3.2. Linking design

The best design to put two sets of items on the same scale would involve all items being administered to all persons; however, we wanted to ensure that no respondent’s test exceeded 15 items, so we had to limit the number of items on a common test comprising all items (i.e., human and GPT-generated). Specifically, we connected two types of items (human-designed textbook items and ChatGPT-generated items) using link forms. Four link forms were established to keep the test length similar across all test forms, ensuring respondents do not experience test fatigue. Another merit of using multiple link forms rather than one is that it allows us to double-check the quality of the link and measurement invariance. In total, there are six forms (OpenStax form, ChatGPT form, and four link forms). Each respondent will be randomly assigned one form from the available six. The assignment of items to the four link forms is based on item property information gained from initial calibrations; namely, we rank all items by difficulty within the original form and form an ‘easy’ test and a ‘hard’ test. Through this assignment, we have a good grasp of the overall difficulty for each form. The test design of our study is shown in Fig. 2. Even though the method does not require equivalent samples, the sampling and resampling process from a crowd-sourcing platform (which we will introduce shortly in the next section) can be safely assumed equivalent, in part due to the recruitment criteria applied. This assumption ensures the estimation process is unbiased.

3.3. Data collection

We recruited respondents via Prolific, a popular crowd-sourcing platform, and utilized an open-source tutoring platform, OATutor (Pardos et al., 2023), to deliver questions and collect responses, as this platform had already transcribed OpenStax College Algebra question content for data collection. OATutor was run with all correctness feedback and other tutoring facilities turned off (i.e., “test mode”) for this study, and the

[4] https://openstax.org/details/books/college-algebra-2e.

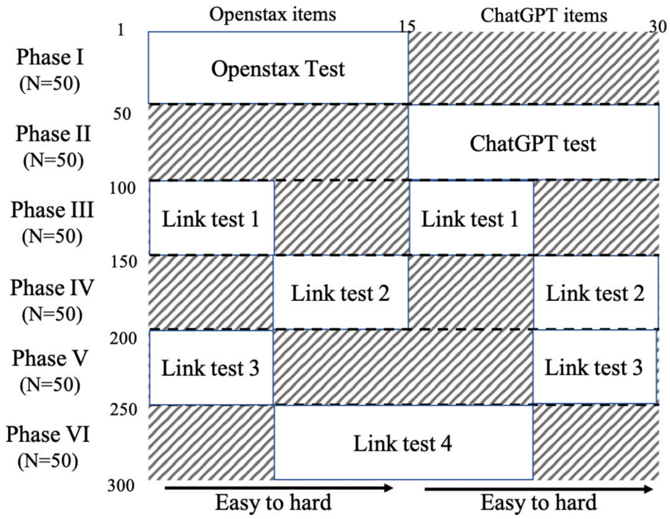

Fig. 2. Our Test design using common item equating.

order of the questions was randomized per user. To ensure the quality of answers, screening tests and quality checks were performed. All respondents who spent less than five minutes on the test and those who completed less than 70% of the questions were excluded from further analysis. Each test form comprised around 15 items focused on a specific topic in college algebra; therefore, the overall test time was estimated to be under 30 minutes. We restricted all respondents to current college students in the United States to ensure they were familiar with the prerequisites of the test domain. There were six forms randomly assigned to respondents, with each respondent taking no more than one form. These forms were distributed to a minimum of 50 participants each. As a result, every individual item accumulated at least 150 valid responses, fulfilling the basic requirement for the Rasch analysis (Wang & Chen, 2005).

3.4. Item response analysis

In the field of measurement, item calibration refers to the empirical process performed on the measurement tool to establish the relation between the scale of the tool and the latent ability it tries to measure (Mari et al., 2023). We applied Item Response Theory (IRT) for item calibration. IRT models the response of each examinee of a given ability to each item in the test as follows:

$$P(X_{ij} = 1|\theta_i, b_j) = \frac{exp(\theta_i - b_j)}{1 + exp(\theta_i - b_j)}$$

This is referred to as the 1-PL model, also commonly known as the Rasch analysis, where the probability of respondent i getting item j correct is defined by ability (θ) and item difficulty (b). Item difficulty varies by item; it's the location where respondents at this level have a 50% chance of getting the item correct. The item information is maximized when the item location is the same as the respondent's location (Wilson, 2023). Therefore, the test is effective when the range of item parameters can well cover the range of respondents' ability distributions. A useful tool to visually see if the test fulfills this requirement is to draw the Wright map (Wilson, 2023), a figure holistically showing item location (difficulty) and person location (ability) at the same time.

Another widely used IRT model is the two-parameter logistic model, as follows:

$$P(X_{ij} = 1|\theta_i, a_j, b_j) = \frac{exp(a_j(\theta_i - b_j))}{1 + exp(a_j(\theta_i - b_j))}$$

In addition to the item difficulty b_j, this model also estimates item discrimination a_j. The discrimination parameter is allowed to vary between items, representing how steeply the rate of success of individuals varies with their ability. Therefore, the higher the discrimination parameter, the better the item will be at discriminating respondents. A well-constructed set of items should have an item discrimination distribution centered around 1 but skewed to the right, which suggests moderately high (>1) discrimination. Empirically, item difficulty does not differ significantly between the Rasch analysis and the 2PL model.

The validity of new items will be examined through the process suggested by previous literature (Mislevy et al., 2003; Wilson, 2023). We calibrated all forms together, which is referred to as the concurrent calibration method (Wingersky & Lord, 1984), because it is more efficient, providing smaller standard errors and involving fewer assumptions than separate calibration procedures. Since different groups in the study were assumed to be equivalent, the estimation with incomplete/non-randomly missing data should not be subject to bias (DeMars, 2002). Items were generated for a formative purpose, meaning they would be mainly used in textbooks as homework or reflection questions. The formative nature allows for making inferences about student ability based on ongoing assessments (DiCerbo et al., 2017). The ideal item should be of moderate difficulty and able to distinguish between low and high proficiency so that a student can self-evaluate and receive diagnostic information from those items.

Response time is one of the most studied features of process data. From an assessment perspective, analyzing response time can help distinguish between respondents' solution behavior and rapid-guessing behavior, thus improving the estimation of student ability (Meyer, 2010). From a formative learning perspective, response time is relevant to the efficiency of learning since shorter response times may eventually lead to a reduced study time.

To compare the quality between two sets of items, we divided the process using Item Response Analysis into different stages:

1. A descriptive analysis was carried out to show differences in item parameters. Specifically, WrightMap allows us to compare the relative difficulty coverage of item sets across the two generation methods. The discrimination parameters were also compared.
2. An unpaired t-test was applied to check whether the item difficulty and item discrimination differ significantly between the two generation methods.
3. A t-test was carried out on the response time for the two sets of items, aiming to check whether there are significant differences in the time respondents spent on each item or each response.

3.5. Dimensionality

Another integral part of the item response analysis is the dimensionality check. Before any test is issued, the dimensionality check can serve as evidence for the internal structure of what the test aims to measure (Wilson, 2023). In our case, our test revolves around our specified learning objective. For a unidimensional construct, multidimensionality (which may be caused by multiple factors) will likely result in poor item fit in the test analysis. This will cause differential item functioning (Ackerman, 1992) and also harm the reliability of the test. Extreme violations of unidimensionality will lead to reconsideration of the rationale for the underlying construct. Without testing for unidimensionality, interpreting the test score on the intended dimension is also risky (Ziegler & Hagemann, 2015). There are multiple indices that can be used for assessing the unidimensionality of tests (Hattie, 1985); some researchers also phrase the problem as assessing the reliability of the test. Therefore, in our study, we use a unidimensionality check using indices based on tetrachoric correlation between items (Kubinger, 2003), whose confidence interval is established by Monte Carlo Simulation (Drasgow & Lissak, 1983). Both sets of items will be checked for their respective dimensionality.

Moreover, to examine whether the two tests load onto the same dimension, we calculated the Cronbach's Alpha for the whole test and each separate test as reliability measures. It is often referred to as internal consistency reliability, the mean of all possible split-half reliabilities (corrected for test length). We compare the whole test reliability and individual test reliability to draw conclusions on whether the addition of the new items impacts the unidimensionality of the original item bank.

In summary, to draw conclusions on the dimensionality evidence of the new items and how they add to the original dimensionality of the textbook questions, we divided the analysis into two stages:

1. A unidimensionality check will be conducted on the OpenStax items and ChatGPT-generated items. Results will determine whether they violate the unidimensionality assumptions.
2. Cronbach's Alpha will be computed for each set of items (i.e., OpenStax or ChatGPT-generated) and for a combined test. A comparison between the individual sets and the combined set (with correction for length) will be used to draw the final conclusion.

3.6. Item similarity

To perform a text similarity analysis between OpenStax and ChatGPT-generated questions, we generated embeddings for ChatGPT questions and three adjacent OpenStax lessons within the same chapter: Lesson 2.1 (The Rectangular Coordinate Systems and Graphs), Lesson 2.2 (Linear Equations in One Variable), and Lesson 2.3 (Models and Applications). We generated these embeddings using Sentence-BERT (Li et al., 2021). We employed Principal Component Analysis (PCA) for dimensionality reduction (i.e., to reduce the vectors to 2D). Additionally, we determined the average cosine similarity between the sets of OpenStax questions and the ChatGPT questions.

To compare the scope of OpenStax and ChatGPT-generated questions, we analyzed content coverage across the three adjacent OpenStax lessons (2.1, 2.2, and 2.3). We focused on comparing ChatGPT-generated questions for Lesson 2.2 with OpenStax questions for the same lesson. We then assessed how this comparison related to the adjacent lessons (2.1 and 2.3) to determine if ChatGPT-generated questions aligned closely with their intended lesson or overlapped significantly with adjacent ones. Our approach involved tagging the questions with the most relevant learning objectives from the OpenStax College Algebra taxonomy, allowing us to evaluate the similarity in scope based on the overlap of learning objectives covered. For the tagging process, we employed a fine-tuned GPT-3.5 model, which demonstrated high agreement with human experts, averaging 90% in this specific task (Kwak & Pardos, 2024). The model was fine-tuned using OpenStax College Algebra questions to predict the most aligned learning objectives from the OpenStax College Algebra taxonomy. Employing this fine-tuned model ensured consistency in tagging across all question sets. To quantify the similarity between learning objectives tagged for ChatGPT-generated questions and those in different OpenStax sections, we calculated weighted embeddings for the learning objectives tagged for each OpenStax and ChatGPT-generated question set. Weighted embeddings are vector representations of learning objectives that account for their frequency in the question set. We then determined the cosine similarity between these weighted embeddings to measure the overall similarity in learning objective coverage.

4. Results

Out of the original 20 ChatGPT questions, two failed quality checks because they were deemed unsolvable, and a random set of 15 from the remaining 18 were selected for the study. Our study with recruited respondents consisted of six phases. Phase 1 consisted of 15 OpenStax items, Phase 2 had 15 ChatGPT items, Phase 3 consisted of the 8 easiest OpenStax items with the 8 easiest ChatGPT items, and Phase 4 contained the 7 hardest OpenStax items with the 7 hardest ChatGPT items. Phase 5 consisted of the 8 easiest OpenStax items with the 7 hardest ChatGPT items, and Phase 6 contained the 7 hardest OpenStax items with the 8 easiest ChatGPT items.

After the six-phase study was conducted, a total of 400 respondents were recruited via Prolific (55 for Phase 1, 60 for Phase 2, 62 for Phase 3, 71 for Phase 4, 74 for Phase 5, and 78 for Phase 6). After applying the exclusion criteria, which required a minimum of 5 minutes spent and 70% of questions attempted, the sample size was reduced to 316 respondents (47 for Phase 1, 52 for Phase 2, 56 for Phase 3, 52 for Phase 4, 53 for Phase 5, and 56 for Phase 6), surpassing our initial target of 300 respondents. Attrition rates between the OpenStax items and ChatGPT items were similar (15% for OpenStax and 13% for ChatGPT). The accuracy for each of the phases was 54% (Phase 1), 52% (Phase 2), 59% (Phase 3), 42% (Phase 4), 50% (Phase 5), and 55% (Phase 6). The relatively low accuracy for Phase 4 was expected because it contained all the "hard" questions, while other phases consisted of either a mix of "easy" and "hard" items (Phases 1, 2, 5, 6) or purely "easy" items (Phase 3). Out of all the items, one item from the pool of ChatGPT items had to be removed from further analysis since no respondent was able to provide a correct response. Having an item with a zero percent correct response rate makes IRT analysis infeasible, necessitating the exclusion of this item.

4.1. Item parameter estimates

The item difficulty parameter and respondent ability estimate were generated through the Rasch analysis, both mapping onto the logit scale. The upper portions of the WrightMaps (Fig. 3) show the respondents' ability distributions. The bottom portion plots the item difficulty estimates, which refer to the location where respondents have a 50% chance of getting a particular item correct. In Fig. 3 (left), we can see that difficulties for both item pools range from [-2, 2], while the OpenStax pool is more spread out in the [-2, -1] range. In general, ChatGPT can do a similarly effective job in evaluating the target population (i.e., college students). Overall, ChatGPT-generated items are slightly more difficult than OpenStax on a logit scale (t = *-0.80*, p = *0.44*), but the difference of 0.05 logit is considered almost negligible.

To generate the item discrimination parameter, we applied a two-parameter logistic model. As mentioned earlier, higher discrimination parameters better differentiate the ability levels of respondents based on the correctness of their answers (Masters, 1988). The average discrimination of ChatGPT questions (1.69) is higher compared to OpenStax items (1.26), indicating that ChatGPT questions do a better job of discriminating respondents. However, the difference in discrimination between the two groups of items was not statistically significant (t = *-1.40*, p = *0.17*), partly due to the relatively small sample size (N = 15).

4.2. Response time analysis

In our study, the mean response time for correct responses is 78.07 seconds for OpenStax questions and 70.69 seconds for ChatGPT questions. The mean response time for incorrect responses is 71.64 seconds for OpenStax questions and 78.30 seconds for ChatGPT questions. This indicates that generally ChatGPT questions have a shorter response time when answered correctly, yet a longer response time when answered incorrectly. However, no statistically significant difference in response time was detected while comparing OpenStax and ChatGPT-generated questions, both on the item level (t = *-0.15*, p = *0.88*) and on the response level (t = *-0.33*, p = *0.73*). Fig. 4 shows the density plot of response times for both types of responses and highlights how the two distributions are very similar to each other.

4.3. Dimensionality check

In the current study, we used a unidimensionality check (Kubinger, 2003) whose confidence interval is established by Monte Carlo simulation (Drasgow & Lissak, 1983). Results show that the OpenStax questions

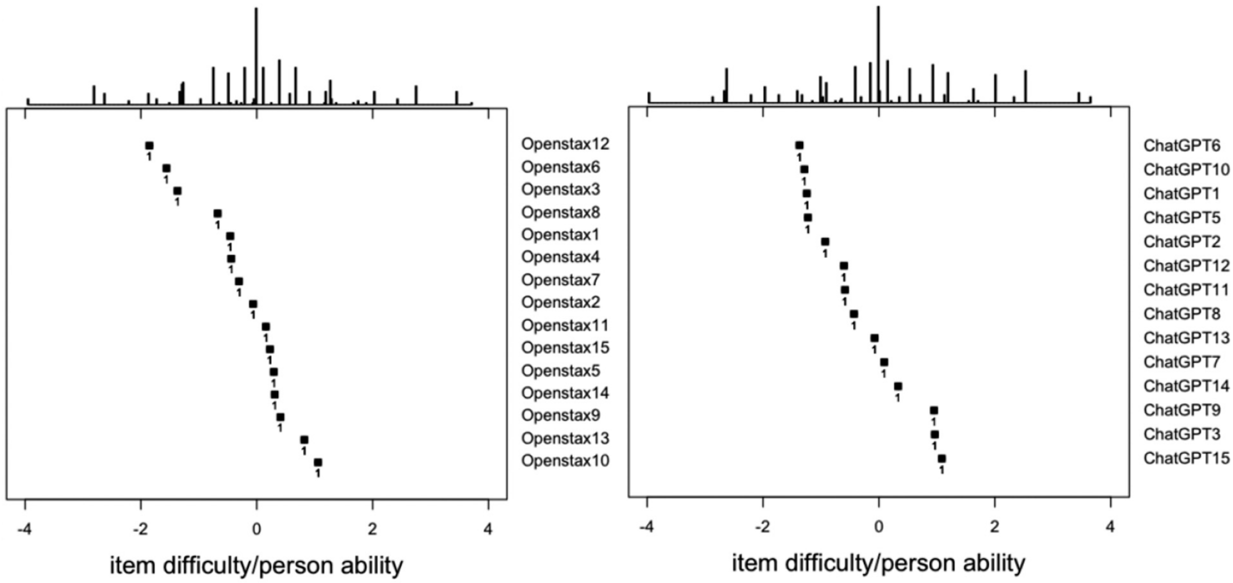

Fig. 3. WrightMap for Openstax items (left) and ChatGPT items (right).

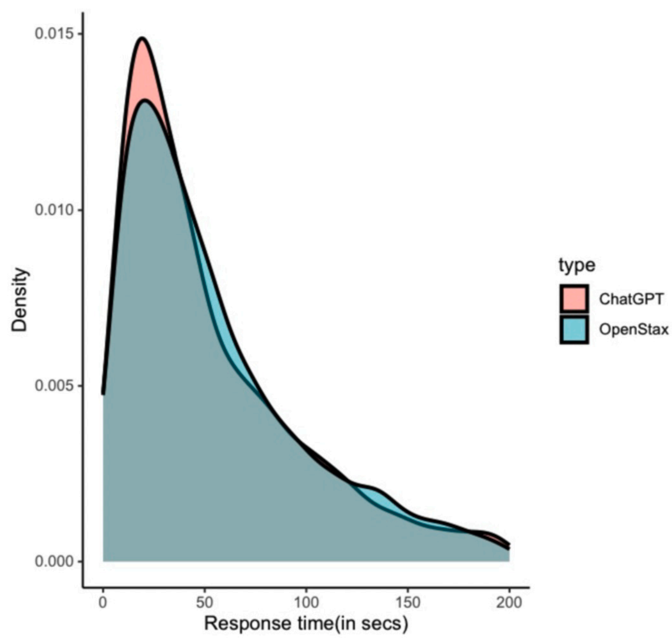

Fig. 4. Density plot of response time.

did not violate the unidimensionality assumption ($p = 0.52$), meaning the OpenStax textbook questions are well aligned with the learning objective. Similarly, the ChatGPT questions did not violate the unidimensionality check as well ($p = 0.08$).

Furthermore, we computed the Cronbach's alpha for the complete test combining items from both pools (N = 30), and the index is 0.88. For the OpenStax scale only (N = 15), the index was 0.83. Since Cronbach's alpha is sensitive to test length, we applied the Spearman-Brown formula to correct for test length; the expected alpha after expansion is 0.91. Therefore, the inclusion of the ChatGPT items has a negative effect on the overall reliability of the test, yet not detrimental, because the reliability after the inclusion (0.88) still lies in the acceptable range of Cronbach's alpha (0.70 to 0.95) (Tavakol & Dennick, 2011).

4.4. NLP-based item similarity analysis

A PCA visualization of the question embeddings is shown in Fig. 5. In the plot, OpenStax 2.3 forms a fairly distinct cluster towards the right side, indicating it has unique features compared to the other sets of questions. The points for ChatGPT and OpenStax 2.2 are interspersed with each other and located primarily on the left side of the plot, suggesting that these sets share more common features. OpenStax 2.1 questions are also located on the left side, but not as closely clustered with the ChatGPT and OpenStax 2.2 questions. Thus, the visualization suggests a significant overlap in the semantic features of the ChatGPT and OpenStax 2.2 questions compared to their relationship with OpenStax 2.1 and OpenStax 2.3 questions.

Average cosine similarities between the sets of OpenStax questions and the ChatGPT questions are shown in Table 1. Considering that the ChatGPT-generated questions have the highest cosine similarity with OpenStax 2.2, this supports the idea that they have the highest alignment.

Further, our analysis of the overlap of learning objectives among the lessons shows that ChatGPT-generated questions have the most similar coverage with OpenStax Lesson 2.2. Out of the 10 learning objectives identified, ChatGPT-generated questions covered five, as did OpenStax Lessons 2.1 and 2.2, while Lesson 2.3 covered two. The overlap in learning objectives was highest between ChatGPT and Lesson 2.2 (four objectives), followed by Lesson 2.1 (two objectives), with no overlap for Lesson 2.3. See Table 2.

Cosine similarity analysis further supports these findings as seen in Table 3. The learning objectives that ChatGPT-generated questions cover showed the highest similarity with those of OpenStax Lesson 2.2 (0.9666), compared to Lesson 2.1 (0.6859) and Lesson 2.3 (0.6153). This high cosine similarity value for Lesson 2.2 indicates that ChatGPT-generated questions align most closely with the intended learning objectives of this lesson.

5. Discussion and conclusion

Item generation using LLMs is a method that can be used to scale the item development process and produce the large numbers of items needed to support Intelligent Tutoring Systems and other formative assessment systems. In this study, the aim of our research was to compare the quality of ChatGPT and human-generated items regarding the psy-

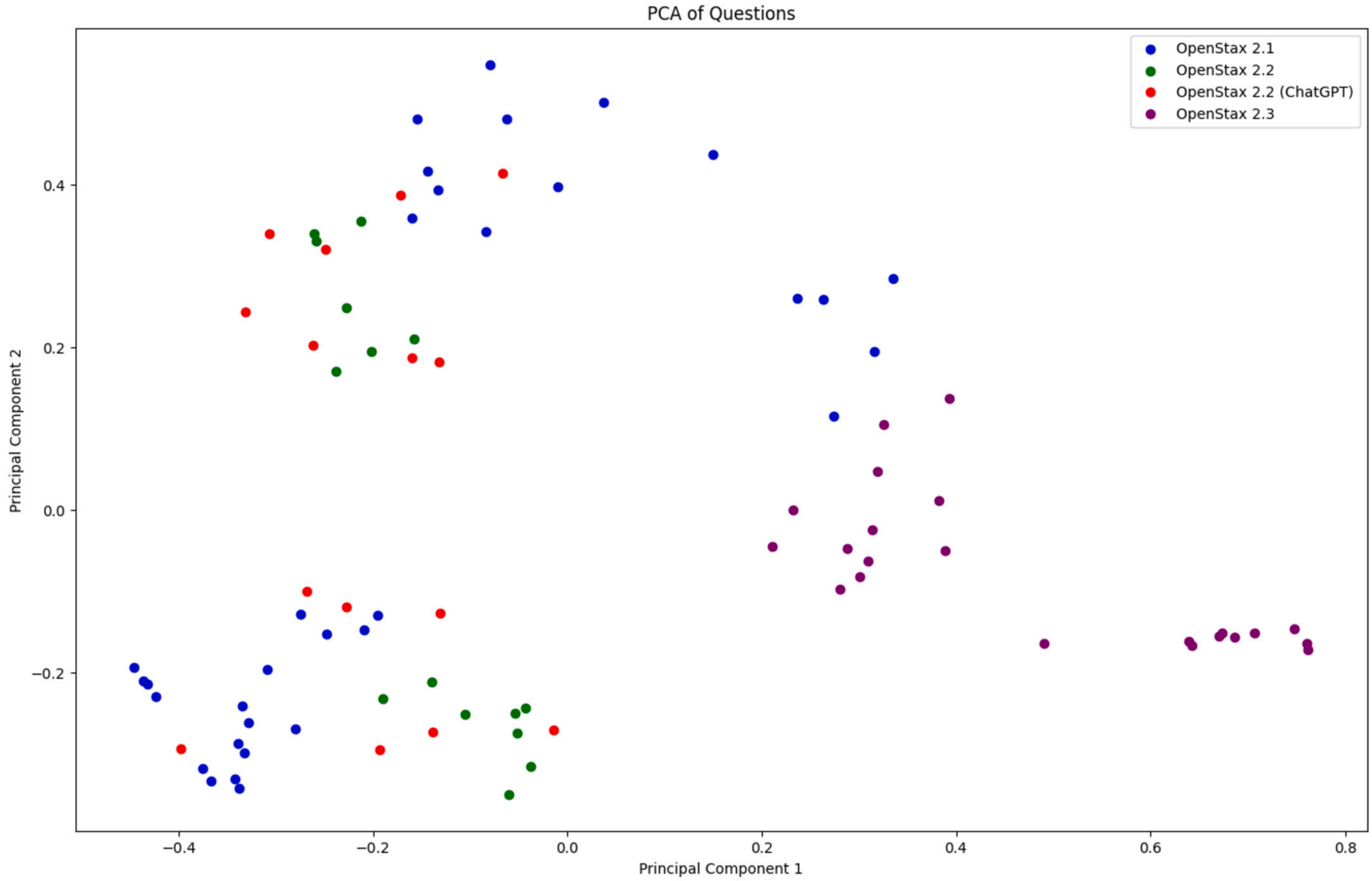

Fig. 5. PCA visualization of question embeddings.

Table 1
Average cosine similarities between question sets.

	ChatGPT	OpenStax 2.1	OpenStax 2.2	OpenStax 2.3
ChatGPT	1.0000	0.3945	0.4439	0.1613
OpenStax 2.1	-	1.0000	0.3582	0.1706
OpenStax 2.2	-	-	1.0000	0.1646
OpenStax 2.3	-	-	-	1.0000

chometric properties for each set of items. The results of our study showed that ChatGPT-generated questions have comparable psychometric properties when compared with gold standard, human-authored textbook questions in College Algebra. Specifically, we found no statistically reliable difference in the difficulty and discrimination parameters between the two. Based on our preliminary analysis, ChatGPT, when given the appropriate prompt, can generate items with locations equally spaced within the ability distribution of respondents and even generate items with higher discrimination power.

While item parameters did not differ, we wanted to further investigate if the subject matter of the generated questions differed. Through PCA visualization of vectorized question content, we found greater overlap in the semantic features of the ChatGPT questions with OpenStax Lesson 2.2 compared to adjacent OpenStax lessons. Additionally, after labeling the learning objectives of generated and existing content, the distribution of learning objectives associated with ChatGPT-generated items had the highest similarity to OpenStax Lesson 2.2 (i.e., Pearson correlation of 0.966) compared to Lesson 2.1 (0.6859) or Lesson 2.3 (0.6153). Considering that the ChatGPT questions were generated from Lesson 2.2, this corroborates that ChatGPT is able to generate questions that cover the intended topic. Since ChatGPT was able to produce high-quality items with aligned content, this suggests LLMs can be used as a useful tool for the development of assessment items, plausibly reducing both time and effort.

Worth discussing is that out of the 15 selected ChatGPT questions, one had to be eliminated from analysis due to a 0% accuracy rate, as is customary in IRT analysis (Wilson, 2023). The particular item required solving quadratic equations, which is not within the scope of the lesson and requires a higher level of ability. This phenomenon indicates that not all ChatGPT items have appropriate qualities. In this study, we used a 3-point quality check criterion to evaluate each ChatGPT-generated question, resulting in a 90% pass rate. These findings highlight the importance of manual checks by subject matter experts after items are automatically generated. Thus, involving both LLMs and humans is necessary to ensure quick, accurate, and robust generation of items.

The study had several limitations. Our current item generation was based on only a single lesson in the OpenStax College Algebra textbook. Due to this, we do not know if the results hold for other domains or levels of mathematics. Another unstudied area is how the number of items will influence the results, especially the distribution of ChatGPT-generated item difficulties. It is uncertain whether the distribution will be more spread out or centered when more items are produced. The current analysis was also restricted in scope, evaluating items based solely on their classical psychometric properties. Future work may also focus on other formative assessment concerns, such as the degree to which the generated items cover the source material concerning concepts and learning objectives.

CRediT authorship contribution statement

Shreya Bhandari: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yunting Liu:** Writing – review

Table 2
Number of problems tagged with each learning objective.

Learning Objective	ChatGPT	OpenStax 2.1	OpenStax 2.2	OpenStax 2.3
Determining whether graphs of lines are parallel or perpendicular	0	0	1 (6.67%)	0
Finding a linear equation	7 (46.67%)	3 (9.09%)	4 (26.67%)	0
Finding x intercepts and y intercepts	1 (6.67%)	15 (45.45%)	0	0
Setting up a linear equation to solve a real world application	0	0	0	16 (72.73%)
Solving a rational equation	2 (13.33%)	0	5 (33.33%)	0
Solving linear equations in one variable	2 (13.33%)	0	3 (20%)	0
Using a formula to solve a real world application	0	1 (3.03%)	0	6 (27.27%)
Using the distance formula	0	9 (27.27%)	0	0
Using the midpoint formula	0	5 (15.15%)	0	0
Writing the equation of lines parallel or perpendicular to a given line	3 (20%)	0	2 (13.33%)	0

Table 3
Average cosine similarities based on learning objectives tagged.

	ChatGPT	OpenStax 2.1	OpenStax 2.2	OpenStax 2.3
ChatGPT	1.0000	0.6859	0.9666	0.6153
OpenStax 2.1	-	1.0000	0.6300	0.3928
OpenStax 2.2	-	-	1.0000	0.6397
OpenStax 2.3	-	-	-	1.0000

& editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yerin Kwak:** Formal analysis, Writing – review & editing. **Zachary A. Pardos:** Formal analysis, Writing – review & editing, Conceptualization, Methodology, Validation, Resources, Data curation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the UC Berkeley Peder Sather Center (Award #50504) and the Vice Provost of Undergraduate Education's Micro Grant Program (Award #39874) for providing financial support for this work. This study was approved by the UC Berkeley Committee for the Protection of Human Subjects under IRB Protocol #2022-12-15943.

Appendix A

ChatGPT lesson 2.2

1. Solve for x: $5x + 10 = 0$.
2. Solve for x: $3x - 8 = 0$.
3. If $\frac{x}{3} + \frac{2x}{5} = 6$, find x.
4. Solve the equation $\frac{3x-5}{x^2-4} = \frac{2x+3}{x+2}$ and check for any extraneous solutions.
5. Given the points $A(2,4)$ and $B(4,8)$, calculate the slope of the line passing through these points.
6. Determine the slope of the equation $y = 3x + 5$.
7. Determine the y-intercept of the equation $y = 3x + 5$.
8. What is the equation of the line with a slope of 4 and passing through the point $(1,3)$?
9. Using the points $P(-1,-2)$ and $Q(3,4)$, derive the equation of the line passing through these points.
10. Convert the equation $y - 2x = 5$ into standard form.
11. Write the equation of a horizontal line passing through the point $(4,-3)$.
12. What is the equation of the vertical line that passes through $(7,6)$?
13. If one line has the equation $y = 2x + 1$, provide the equation of a line parallel to it and passing through the point $(3,2)$.
14. For the line $y = -3x - 4$, write the equation of another line that's parallel to it and passes through the point $(-1,5)$.
15. If a line has the equation $y = 4x + 2$, what is the equation of a line perpendicular to it and passing through $(0,3)$?

OpenStax lesson 2.2

1. Given $m = 4$, find the equation of the line in slope-intercept form passing through the point $(2,5)$.
2. Find the slope of a line that passes through the points $(2,-1)$ and $(-5,3)$.
3. Solve the following equation: $2x + 1 = -9$.
4. State whether the two lines are parallel, perpendicular, or neither: $3y = -4x + 3$ and $3x - 4y = 8$.
5. Write the equation of a line parallel to $5x + 3y = 1$ and passing through the point $(3,5)$.
6. Solve the following equation: $2x + 7 = 19$.
7. Find the slope of a line that passes through the points $(-2,6)$ and $(1,4)$.
8. Find the equation of the line passing through the given points: $(-5,2)$ and $(2,2)$.
9. Find the equation of the line parallel to $5x = 7 + y$ and passing through the point $(-1,-2)$.
10. Solve the rational equation: $\frac{7}{2x} - \frac{5}{3x} = \frac{22}{3}$.
11. Solve the following rational equations and state the excluded values: $\frac{3}{x-6} = \frac{5}{x}$.
12. Solve the equation in one variable: $4x - 3 = 5$.
13. Solve the following rational equation: $\frac{2}{x} - \frac{3}{2} = \frac{7}{2x}$.
14. For the following exercises, solve the equation for x. State all x-values that are excluded from the solution set: $\frac{1}{x} = \frac{1}{5} + \frac{3}{2x}$.
15. For the following exercises, solve the equation for x. State all x-values that are excluded from the solution set: $2 - \frac{3}{x+4} = \frac{x+2}{x+4}$.

References

Abramson, J. (2021). *College algebra 2e*. Houston, Texas: OpenStax.

Ackerman, T. A. (1992). A didactic explanation of item bias, item impact, and item validity from a multidimensional perspective. *Journal of Educational Measurement*, *29*(1), 67–91.

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., et al. (2021). On the opportunities and risks of foundation models. arXiv preprint, arXiv:2108.07258.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, Girish, Askell, Amanda, et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, *33*, 1877–1901.

Dhawaleswar, Rao C. H., & Saha, Sujan Kumar (2023). Generation of multiple-choice questions from textbook contents of school-level subjects. *IEEE Transactions on Learning Technologies*, *16*(1), 40–52.

Colvin, K. F., Champaign, J., Liu, A., Zhou, Q., Fredericks, C., & Pritchard, D. E. (2014). Learning in an introductory physics mooc: All cohorts learn equally, including an on-campus class. *The International Review of Research in Open and Distributed Learning*, *15*(4).

Damanik, N. H., & Zainil, Y. (2019). The analysis of reading comprehension questions in English textbook by using higher order thinking skill at grade x of sman 2 padang. *Journal of English Language Teaching*, *8*(1), 249–258.

DeMars, C. (2002). Incomplete data and item parameter estimates under jmle and mml estimation. *Applied Measurement in Education*, *15*(1), 15–31.

DiCerbo, K., Shute, V., & Kim, Y. J. (2017). The future of assessment in technology rich environments: Psychometric considerations. In *Learning, design, and technology: An international compendium of theory, research, practice, and policy* (pp. 1–21).

Drasgow, F., & Lissak, R. I. (1983). Modified parallel analysis: A procedure for examining the latent dimensionality of dichotomously scored item responses. *Journal of Applied Psychology*, *68*(3), 363.

Griffith, S., Subramanian, K., Scholz, J., Isbell, C. L., & Thomaz, A. L. (2013). Policy shaping: Integrating human feedback with reinforcement learning. *Advances in Neural Information Processing Systems*, *26*.

Hattie, J. (1985). Methodology review: Assessing unidimensionality of tests and ltenls. *Applied Psychological Measurement*, *9*(2), 139–164.

Irvine, S. H., & Kyllonen, P. C. (2013). *Item generation for test development*. Routledge.

Joo, S.-H., Lee, P., & Stark, S. (2017). Evaluating anchor-item designs for concurrent calibration with the ggum. *Applied Psychological Measurement*, *41*(2), 83–96.

Keller, S. U. (2021). *Automatic generation of word problems for academic education via natural language processing (nlp)*.

Kubinger, K. D. (2003). On artificial results due to using factor analysis for dichotomous variables. *Psychology Science*, *45*(1), 106–110.

Kurdi, G., Leo, J., Parsia, B., Sattler, U., & Al-Emari, S. (2020). A systematic review of automatic question generation for educational purposes. *International Journal of Artificial Intelligence in Education*, *30*, 121–204.

Kwak, Y., & Pardos, Z. A. (2024). Bridging large language model disparities: Skill tagging of multilingual educational content. *British Journal of Educational Technology*.

Lane, S., Raymond, M. R., Haladyna, T. M., & Downing, S. M. (2015). Test development process. In *Handbook of test development* (pp. 3–18). Routledge.

Li, Z., Ren, C., Li, X., & Pardos, Z. A. (2021). Learning skill equivalencies across platform taxonomies. In *LAK21: 11th international learning analytics and knowledge conference* (pp. 354–363).

Liu, Y., Bhandari, S., & Pardos, Z. A. (2024). Leveraging llm-respondents for item evaluation: A psychometric analysis. arXiv preprint, arXiv:2407.10899.

Lu, X., & Wang, X. (2024). Generative students: Using llm-simulated student profiles to support question item evaluation. In *Proceedings of the eleventh ACM conference on learning@ scale* (pp. 16–27).

Mari, L., Wilson, M., & Maul, A. (2023). *Measurement across the sciences: Developing a shared concept system for measurement*. Springer Nature.

Masters, G. N. (1988). Item discrimination: When more is worse. *Journal of Educational Measurement*, *25*(1), 15–29.

McHorney, C. A. (2002). Use of item response theory to link 3 modules of functional status items from the asset and health dynamics among the oldest old study. *Archives of Physical Medicine and Rehabilitation*, *83*(3), 383–394.

Meyer, J. P. (2010). A mixture rasch model with item response time components. *Applied Psychological Measurement*, *34*(7), 521–538.

Meyer, J. P., & Zhu, S. (2013). Fair and equitable measurement of student learning in moocs: An introduction to item response theory, scale linking, and score equating. *Research & Practice in Assessment*, *8*, 26–39.

Mishra, R. K. (2015). Mapping the knowledge topography: A critical appraisal of geography textbook questions. *International Research in Geographical and Environmental Education*, *24*(2), 118–130.

Mislevy, R. J., Steinberg, L. S., & Almond, R. G. (2003). Focus article: On the structure of educational assessments. *Measurement: Interdisciplinary Research and Perspectives*, *1*(1), 3–62.

Moore, S., Nguyen, H. A., Chen, T., & Stamper, J. (2023). Assessing the quality of multiple-choice questions using gpt-4 and rule-based methods. In *European conference on technology enhanced learning* (pp. 229–245). Springer.

Nguyen, H. A., Bhat, S., Moore, S., Bier, N., & Stamper, J. (2022). Towards generalized methods for automatic question generation in educational domains. In *European conference on technology enhanced learning* (pp. 272–284). Springer.

Nguyen, H. A., Stec, H., Hou, X., Di, S., & McLaren, B. M. (2023). Evaluating chatgpt's decimal skills and feedback generation in a digital learning game. In *European conference on technology enhanced learning* (pp. 278–293). Springer.

Onal, S., & Kulavuz-Onal, D. (2024). A cross-disciplinary examination of the instructional uses of chatgpt in higher education. *Journal of Educational Technology Systems*, *52*(3), 301–324.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. (2022). Training language models to follow instructions with human feedback. arXiv preprint, arXiv:2203.02155.

Overman, M., Vermunt, J. D., Meijer, P. C., Bulte, A. M., & Brekelmans, M. (2013). Textbook questions in context-based and traditional chemistry curricula analysed from a content perspective and a learning activities perspective. *International Journal of Science Education*, *35*(17), 2954–2978.

Pardos, Z. A., & Bhandari, S. (2023). Learning gain differences between chatgpt and human tutor generated algebra hints. arXiv preprint, arXiv:2302.06871.

Pardos, Z. A., Tang, M., Anastasopoulos, I., Sheel, S. K., & Zhang, E. (2023). Oatutor: An open-source adaptive tutoring system and curated content library for learning sciences research. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. New York, NY, USA: Association for Computing Machinery.

Radford, A., Narasimhan, K., Salimans, T., Sutskever, I., et al. (2018). *Improving language understanding by generative pre-training*.

Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al. (2019). *Language models are unsupervised multitask learners*.

Sayin, A., & Gierl, M. (2024). Using openai gpt to generate reading comprehension items. *Educational Measurement: Issues and Practice*, *43*(1), 5–18.

Shridhar, K., Macina, J., El-Assady, M., Sinha, T., Kapur, M., & Sachan, M. (2022). *Automatic generation of socratic subquestions for teaching math word problems*.

Sireci, S. G., & Zenisky, A. L. (2011). Innovative item formats in computer-based testing: In pursuit of improved construct representation. In *Handbook of test development* (pp. 343–362). Routledge.

Tan, B., Armoush, N., Mazzullo, E., Bulut, O., & Gierl, M. J. (2024). A review of automatic item generation techniques leveraging large language models.

Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. *International Journal of Medical Education*, *2*, 53.

Vale, C. D. (1986). Linking item parameters onto a common scale. *Applied Psychological Measurement*, *10*(4), 333–344.

van der Linden, W. J., & Barrett, M. D. (2016). Linking item response model parameters. *Psychometrika*, *81*(3), 650–673.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.

von Davier, M., Yamamoto, K., Shin, H. J., Chen, H., Khorramdel, L., Weeks, J., Davis, S., Kong, N., & Kandathil, M. (2019). Evaluating item response theory linking and model fit for data from pisa 2000–2012. *Assessment in Education: Principles, Policy & Practice*, *26*(4), 466–488.

Wang, W.-C., & Chen, C.-T. (2005). Item parameter recovery, standard error estimates, and fit statistics of the winsteps program for the family of rasch models. *Educational and Psychological Measurement*, *65*(3), 376–404.

Wilson, M. (2009). Measuring progressions: Assessment structures underlying a learning progression. *Journal of Research in Science Teaching: The Official Journal of the National Association for Research in Science Teaching*, *46*(6), 716–730.

Wilson, M. (2013). Using the concept of a measurement system to characterize measurement models used in psychometrics. *Measurement*, *46*(9), 3766–3774.

Wilson, M. (2023). *Constructing measures: An item response modeling approach*. Taylor & Francis.

Wingersky, M. S., & Lord, F. M. (1984). An investigation of methods for reducing sampling error in certain irt procedures. *Applied Psychological Measurement*, *8*(3), 347–364.

Wright, B. D., & Stone, M. H. (1979). *Best test design*. Mesa Press.

Zhou, Z., Ning, M., Wang, Q., Yao, J., Wang, W., Huang, X., & Huang, K. (2023). *Learning by analogy: Diverse questions generation in math word problem*.

Ziegler, M., & Hagemann, D. (2015). *Testing the unidimensionality of items*.